

HARIHARAN NARLAKANTI

AI Engineer | NLP | LLM Applications | Backend
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SUMMARY

Computer Science graduate focused on AI engineering, with hands-on experience in Python, NLP, and LLM application development. Built RAG pipelines, semantic search systems, and production-oriented FastAPI/backend services, with work spanning transformer fine-tuning, vector retrieval, and containerized deployment.

EDUCATION

Malla Reddy University, Hyderabad

2022 – 2026

Bachelor of Technology – Computer Science & Engineering (AI & ML), CGPA: **8.89 / 10**

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, DBMS, Software Engineering

TECHNICAL SKILLS

AI / ML: Python, PyTorch, Transformers, SentenceTransformers, NLP, RAG, LLM Applications, ChromaDB, LangChain

Backend: FastAPI, REST APIs, Flask, JWT Authentication

Databases: PostgreSQL, MongoDB, MySQL

Tools: Docker, Git, Linux, React, Next.js

EXPERIENCE

Full Stack Development Intern – SmartInternz

May 2025 – Jul 2025

- Built **SocialeX**, a full-stack social networking application on the MERN stack, implementing authenticated REST APIs, JWT-based access control, and MongoDB-backed data models.
- Implemented real-time one-to-one messaging via Socket.IO and enforced server-side authorization across user profiles, posts, and protected resources.

PROJECTS

Saaram – AI-Powered Telugu News Summarization & Speech Platform

GitHub | Live Demo

- Designed a hybrid Telugu NLP pipeline combining morphology-aware TF-IDF, mT5 transformers, and **QLoRA** fine-tuning, with adaptive model routing and resource-aware fallbacks.
- Deployed a containerized FastAPI service exposing **5+ REST endpoints** and evaluated summarization quality on the **1,302-sample XL-Sum Telugu** subset.

ResearchCompass – AI Research Paper Analyzer

GitHub | Live Demo

- Engineered a document intelligence platform on Next.js and FastAPI, using PyMuPDF for parsing and a **RAG** pipeline with ChromaDB and SentenceTransformers for retrieval.
- Integrated Groq-served LLM analysis for structured critiques, cross-paper comparison, and research-readiness scoring.

VoiceVibes AI – Vocal Emotion Analytics Dashboard

GitHub

- Built a full-stack audio analytics application with a fine-tuned **Wav2Vec2** transformer running CPU inference behind a FastAPI backend.
- Implemented a custom browser-side WAV encoder for mono 16-bit PCM audio, visualizing real-time probabilities across **7 emotion classes**.

Smart-FAQ – Semantic Search Platform

GitHub

- Developed a FastAPI and React semantic search service using SentenceTransformers embeddings and cosine-similarity matching against a MongoDB document store.
- Secured admin operations with JWT authentication and added confidence-based fallback handling for low-similarity queries.

ACHIEVEMENTS & RESEARCH

- Best Project Award**, Malla Reddy University – Telugu NLP news summarization platform.
- Co-authored a research paper on Telugu text summarization submitted to **CIS 2026 (Springer LNNS)**.

CERTIFICATIONS

- Machine Learning Specialization – DeepLearning.AI & Stanford University (2026)
- IBM Full Stack Software Developer Professional Certificate (2024)
- AWS Academy Cloud Foundations (2025)
- MongoDB Node.js Developer Path (2025)